Rodrigo Espiritu Berra

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GitHub | LinkedIn | Puebla, Mexico

Professional Experience

AI & Automation Consultant **Ellucian** **Jun 2026 – Present**

- Engineered enterprise AI agent runtimes using AWS Bedrock AgentCore, Strands Agents, MCP, Python, and AWS CDK, enabling modular tool integrations and cloud deployment.
- Developed agent-accessible knowledge retrieval systems for Confluence, Jira, and Box, incorporating metadata processing, document references, and enterprise search.
- Implemented identity-aware Jira integrations with secure user resolution, permission checks, and fail-closed authorization for agent tools.
- Designed an automated AI agent evaluation framework with 24 test cases and repeated executions, identifying three behavioral defects in tool selection and confirmation.
- Investigated MCP integration failures and performance bottlenecks, using AWS observability tools to diagnose latency, errors, and agent resource consumption.
- Improved React/TypeScript dashboard UX and reduced a JavaScript vendor bundle from 801 KB to 239 KB (approximately 70%).

AI & Automation Consultant Intern **Ellucian** **Feb 2026 – May 2026**

- Developed an end-to-end AI onboarding chatbot integrated with Slack using Amazon Bedrock Knowledge Bases, AgentCore, Flows, Guardrails, Lambda, API Gateway, and DynamoDB.
- Engineered Python-based Confluence ingestion pipelines using Atlassian APIs and Selenium, implementing data cleaning, metadata generation, and semantic chunking.
- Optimized knowledge retrieval architecture by adopting S3 Vectors, configuring AWS IAM permissions, and implementing Jenkins CI/CD pipelines.

Software Development Intern **Merkur.ia** **Jun 2025 – Aug 2025**

- Developed LangChain/LangSmith AI agents, improving data visualization and interpretability by 80%.
- Built Python scraping pipelines and a React Native/TypeScript dashboard with Pinecone, improving efficiency by 50%.

Quality and Development Analyst in Generative AI **Outlier AI** **Sep 2024 – Jan 2026**

- Evaluated LLM outputs, optimized system prompts and AI-generated Python/JSON code, and validated APIs against quality standards.

O-Quality & Process Management Intern **T-Systems Mexico** **Apr 2024 – Aug 2024**

- Developed GPT-powered mini-games using Microsoft Power Apps, prompt engineering, and JSON/CSV integration.

Education

Universidad de las Americas Puebla (UDLAP) *Jun 2026*
B.S. in Software Engineering GPA: 3.8

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, C++, Swift, Shell

AI & Agentic Systems: Amazon Bedrock, AgentCore, Strands Agents, MCP, RAG, LangChain, LangSmith, LLM Evaluation.

Languages: Spanish, English

Cloud & DevOps: AWS Lambda, S3, S3 Vectors, DynamoDB, API Gateway, AWS CDK, CloudFormation, IAM, CloudWatch, Jenkins, Docker, Git, Datadog

Frameworks & Tools: React, React Native, Node.js, SwiftUI, Selenium, Flask, OpenCV, Pinecone, REST APIs, Jira, Confluence, Figma

Projects

Driver Fatigue Detection System *2025*
SwiftUI, Python, TensorFlow, OpenCV, MobileNetV2
Developed an iOS fatigue detection system using computer vision, real-time video analysis, and deep learning.

TrustyWeb *2024*
React, TypeScript, Node.js, OpenAI API
Developed an AI-powered Chromium extension for online information analysis and verification.

Certification

AWS Solution Architect (Certification in progress) Google Cloud Computing Foundations Certification (2025)